

**PH.D. QUALIFYING EXAM 2019: STATISTICAL
INFERENCE**

Rules: This is a four hour exam consisting of two problems, both of which should be answered. You may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN “cheat sheet”. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Finally, write your answers clearly and legibly on blank papers, preferably using pens, leaving an impression deep enough for the papers to be scanned. At the end of the exam, separate out the bunches of pages for problems 1 and 2, and hand them over to the proctor. Good luck!

Problem 1

Instructions:

- If you cannot solve/prove one claim completely, provide as many partial details as possible.
- Your final score for this problem = $\min\{100, \text{raw score}\}$.

Part I. (30 points)

- (1) Let $\{P_n\}, \{Q_n\}$ be a sequence of probability measures with P_n, Q_n defined on a measurable space $(\Omega_n, \mathcal{A}_n)$. State the definition for $\{Q_n\}$ to be contiguous to $\{P_n\}$.
- (2) State the definition for an experiment $(P_\theta : \theta \in \Theta \subset \mathbb{R}^k)$ to be local asymptotically normal (LAN) (at θ), and to be uniformly LAN.

Part II. (50 points)

Suppose that the experiment $(P_\theta : \theta \in \Theta \subset \mathbb{R}^k)$ is dominated by a σ -finite measure μ with densities $\{p_\theta : \theta \in \Theta\}$, and is uniformly LAN at θ with invertible Fisher information matrix I_θ . Let $H_n = \sqrt{n}(\Theta - \theta) \rightarrow H$ ¹ for some nonempty and convex and closed H .

Let $\hat{\theta}_n$ be the MLE defined via $\hat{\theta}_n = \arg \max_{\vartheta \in \Theta} \prod_{i=1}^n p_\vartheta(X_i)$ where $X_1, \dots, X_n$ are i.i.d. draws from P_θ . We assume for simplicity that the MLE $\hat{\theta}_n$ is well-defined and $\sqrt{n}$ -consistent: $\sqrt{n}(\hat{\theta}_n - \theta) = O_{P_\theta^n}(1)$. Following the steps to be specified below, show that for any $h \in \mathbb{R}^k$ such that $\theta + h/\sqrt{n} \in \Theta$ for n large enough²,

$$\sqrt{n}(\hat{\theta}_n - \theta) \rightsquigarrow_{P_{\theta+h/\sqrt{n}}^n} I_\theta^{-1/2} \Pi_{I_\theta^{1/2}H} (I_\theta^{1/2} Y_h).$$

Here $Y_h \sim N(h, I_\theta^{-1})$ and for a convex and closed K , Π_K is the projection operator onto K with respect to Euclidean norm, i.e. for any $u \in \mathbb{R}^k$, $\Pi_K(u) \equiv \arg \min_{v \in K} \|u - v\|^2$.

In particular, if $\theta \in \text{int}(\Theta)$, then $H = \mathbb{R}^k$, and we have for any $h \in \mathbb{R}^k$, $\sqrt{n}(\hat{\theta}_n - \theta) \rightsquigarrow_{P_{\theta+h/\sqrt{n}}^n} Y_h$. This recovers the most elementary case $h = 0$.

¹The set convergence is defined as in class: $H_n \rightarrow H$ if and only if for any $h \in H$, there exist a sequence $h_n \in H_n$ such that $h_n \rightarrow h$ in $\mathbb{R}^k$.

²Recall that the weak convergence $Z_n \rightsquigarrow_{P_n} Z$ if and only if for any bounded continuous function f , $E_{P_n} f(Z_n) \rightarrow E f(Z)$.

- Show that for fixed closed set F , for n large,

$$\begin{aligned} P_{\theta+h/\sqrt{n}}^n(\hat{h}_n \in F) &\leq P_{\theta+h/\sqrt{n}}^n \left(\|I_\theta^{-1/2} \mathbb{G}_n \dot{\ell}_\theta - I_\theta^{1/2} (F \cap H_n)\| \right. \\ &\quad \left. \leq \|I_\theta^{-1/2} \mathbb{G}_n \dot{\ell}_\theta - I_\theta^{1/2} H_n\| + o_p(1) \right). \end{aligned}$$

Here $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P)$ is the empirical process, and $\dot{\ell}_\theta$ the first derivative vector of the log-likelihood with respect to θ .

Hint: $\hat{h}_n \in F$ implies that

$$\sup_{h \in F \cap H_n} \mathbb{P}_n \log \frac{f_{\theta+h/\sqrt{n}}}{f_\theta} \geq \sup_{h \in H_n} \mathbb{P}_n \log \frac{f_{\theta+h/\sqrt{n}}}{f_\theta}.$$

Then use LAN.

- Find the limit distribution of $\mathbb{G}_n \dot{\ell}_\theta$ under $P_{\theta+h/\sqrt{n}}^n$ using Le Cam's third lemma.
- Use Portmanteau theorem to conclude.

You may use the following facts without proof:

- *Uniform LAN implies that for any $M > 0$,*

$$\sup_{\|h\| \leq M} \left| n \mathbb{P}_n \log \frac{f_{\theta+h/\sqrt{n}}}{f_\theta} - h^\top \mathbb{G}_n \dot{\ell}_\theta + \frac{1}{2} h^\top I_\theta h \right| \rightarrow_{P_\theta^n} 0.$$

- *Suppose the sets $H_n \rightarrow H$ and the random vectors $X_n \rightsquigarrow X$. Then*
 - $\|X_n - H_n\| \rightsquigarrow \|X - H\|$.
 - *For any closed set F , $\|X_n - H_n \cap F\| \geq \|X_n - H \cap F\| + o_P(1)$.*
 - *For any open set G , $\|X_n - H_n \cap G\| \leq \|X_n - H \cap G\| + o_P(1)$.*

Part III. (50 points)

Suppose that the experiment $(P_\theta : \theta \in \Theta \subset \mathbb{R}^k)$ is dominated by μ , and is uniformly LAN at θ with invertible Fisher information matrix I_θ . Let $\psi : \Theta \rightarrow \mathbb{R}$ be a differentiable functional with derivative (vector) $\dot{\psi}_\theta$. Suppose the map $\theta \mapsto dP_\theta/d\mu$ is sufficiently smooth (in whatever sense you want to make, but be sure to clearly state the conditions you assume). Following the minimax machinery to be specified below, prove that there exists a universal constant $L > 0$ such that

$$\begin{aligned} &\sup_{\tau > 0} \liminf_{n \rightarrow \infty} \inf_{T_n} \sup_{\vartheta \in \Theta : \|\vartheta - \theta\|^2 \leq \tau/n} E_\vartheta \left[\left| \sqrt{n}(\psi(T_n) - \psi(\vartheta)) \right|^2 \right] \\ &\geq L^{-1} \cdot \dot{\psi}_\theta^\top I_\theta^{-1} \dot{\psi}_\theta. \end{aligned}$$

Here E_ϑ means expectation taken under the n -fold product measure $P_\vartheta^{\otimes n}$, i.e. n i.i.d. observations.

- Suppose $\theta_0, \theta_1 \in \Theta$ are such that $|\psi(\theta_0) - \psi(\theta_1)| \geq \gamma_n$, show that

$$\inf_{T_n} \sup_{\vartheta \in \{\theta_0, \theta_1\}} E_\vartheta |\psi(T_n) - \psi(\vartheta)| \geq L^{-1} \gamma_n \cdot \inf_{\phi} \max_{j=0,1} P_{\theta_j}(\phi \neq j).$$

Here the infimum is taken over all tests ϕ taking values in $\{0, 1\}$.

Hint: For any estimator T_n , consider the test $\phi^ = \arg \min_{j=0,1} |\psi(T_n) - \psi(\theta_j)|$.*

- Show that

$$\begin{aligned} \inf_{\phi} \max_{j=0,1} P_{\theta_j}(\phi \neq j) &\geq \frac{1}{2} \inf_{\phi} \left[P_{\theta_0}(\phi \neq 0) + P_{\theta_1}(\phi \neq 1) \right] \\ &= \frac{1}{2} \int (p_0 \wedge p_1) \, d\mu \geq \frac{1}{4} \left(\int \sqrt{p_{\theta_0} p_{\theta_1}} \, d\mu \right)^2 \geq \frac{1}{4} \exp(-D(P_{\theta_0} \| P_{\theta_1})), \end{aligned}$$

where $D(\cdot \| \cdot)$ is the Kullback-Leibler divergence.

- Choose appropriate θ_0, θ_1 , and conclude by calculating $\gamma_n, D(P_{\theta_0} \| P_{\theta_1})$.

The above result shows that the variance lower bound in the convolution theorem cannot be improved—up to an absolute constant—in another local asymptotic minimax sense, without regularity conditions on T_n within the infimum.

Problem 2

Instructions: This problem consists of two parts that are by and large independent of each other, except for the last sub-part 2c. Show the details of your derivation: partial grades will be given if you make some reasonable progress but cannot solve the problem completely. At the same time, please only show the derivation that you think is sensible. Any wrong argument will lead to a deduction of the grade.

PART 1: *Statistical Inference in high-dimensional parametric regression.* This problem investigates high dimensional regression from three angles: (1) point estimation, (2) confidence interval for low dimension objective and (3) support recovery. Consider the high-dimensional sparse linear regression

$$y_i = X_i^\top \beta + \varepsilon_i \quad \text{for } 1 \leq i \leq n$$

where y is the response, X_i is a p -dimensional random vector with mean zero and covariance $\Sigma = \mathbb{E}(X_i X_i^\top)$, $\beta \in \mathbb{R}^p$, $p \geq n$, $\|\beta\|_0 \leq k$ and ε_i 's are i.i.d normal random variables with mean zero and variance σ^2 and independent of X_i . Here $\|\beta\|_0$ denotes the number of non-zero entries in β .

Part 1a. (5 points) Write down the optimization algorithm for Lasso and specify the choice of tuning parameter. Write down the rates of convergence of the following functions of the Lasso estimator $\hat{\beta}$.

- (1) $\frac{1}{n} \|X(\hat{\beta} - \beta)\|_2^2$
- (2) $\|\hat{\beta} - \beta\|_1$

(Note: you do not need to derive the rates of convergence.)

Part 1b. (10 points) Propose an estimator $\widehat{\sigma^2}$ of σ^2 based on the Lasso estimator $\hat{\beta}$ and establish the rate of convergence for $|\widehat{\sigma^2} - \sigma^2|$.

Hint: (1) think about how to estimate σ^2 in the OLS case, (2) you can use the conclusion from Problem 1a to derive the rate of convergence for $|\widehat{\sigma^2} - \sigma^2|$.

Part 1c. (15 points) Propose a debiased estimator of β_1 and write down the algorithm to derive the estimator.

Part 1d. (20 points) Establish the asymptotic normality of the debiased estimator proposed in Problem 1c and then use this limiting distribution to construct confidence intervals for β_1 .

Hint: For the Lasso estimator $\hat{\beta}$, the sample covariance matrix $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top$ and a positive constant C , you can assume the following events to occur with high probability,

- $\|\hat{\beta} - \beta\|_1 \leq C \|\beta\|_0 \sqrt{\frac{\log p}{n}}$.
- $\left| \frac{u^\top \hat{\Sigma} u}{u^\top \Sigma u} - 1 \right| \leq C \sqrt{\frac{\log p}{n}}$ for a given vector $u \in \mathbb{R}^p$.

Part 1e. (15 points) Based on the limiting distribution established in Problem 1d, propose an estimator $\hat{\mathcal{S}}$ of the support set $\mathcal{S} = \{j : \beta_j \neq 0\}$. Justify that $\hat{\mathcal{S}}$ satisfies

$$\mathcal{S}_{\text{strong}} \subseteq \hat{\mathcal{S}} \subseteq \mathcal{S}, \quad \text{where } \mathcal{S} = \{j : \beta_j \neq 0\} \text{ and } \mathcal{S}_{\text{strong}} = \left\{ j : |\beta_j| \geq C \sqrt{\frac{\log p}{n}} \right\}$$

for some positive constant $C > 0$. (Note: you need to be specific about the thresholding level, which depends on the limiting distribution established in Problem 1d.)

PART 2: *Non-parametric modeling and its extension to high dimensions.* We first review several useful nonparametric methods in the univariate case and then discuss the possibility of extending this to the high dimension setting.

Part 2a. (15 points) Assume that data $\{(y_i, t_i)\}_{1 \leq i \leq n}$ are generated independently and identically following the model

$$(0.1) \quad y_i = f(t_i) + \varepsilon_i,$$

where t_i is a random variable with a density function and ε_i is a mean zero random variable independent of t_i . Write down the smoothing spline estimator of $f(\cdot)$ and the local linear estimator of $f(x_0)$ and $f'(x_0)$, where $f'(x_0)$ denotes the derivative of $f(x_0)$.

Part 2b. (10 points) Derive the closed form of the smoothing spline estimator using the basis approach. You can assume the existence of a set of basis functions.

Part 2c. (10 points) A combination of ideas associated with high-dimensional linear regression in part 1 and univariate nonparametric regression in parts 2a and 2b leads to the following high-dimensional sparse additive model:

$$(0.2) \quad y_i = \sum_{j=1}^p f_j(X_{ij}) + \varepsilon_i, \quad \text{for } 1 \leq i \leq n,$$

where $\{f_j\}_{1 \leq j \leq p}$ are unknown functions, $\mathcal{S} = \{j : f_j \neq 0\}$ denotes the set of non-zero functions and ε_i 's are i.i.d mean zero random variables independent of $\{X_{ij}\}_{1 \leq j \leq p}$. Here, the sparsity of the model is captured by the set $\mathcal{S}$, whose cardinality is much smaller than the sample size n and also the dimension p ; in addition, each nonparametric component f_j is assumed to be smooth. Several nonparametric estimators have been proposed in class to account for smoothness in the univariate nonparametric model, such as, the local linear estimator and smoothing spline. Propose an estimator $\{\hat{f}_j\}_{1 \leq j \leq p}$ to account for both the sparsity and smoothness.

Hints: Here are two possible ways that you can think about the solution:

- (1) Regarding the sparse additive model, the proposed method needs to account for both smoothness and also the sparsity. We have accounted for smoothness in the nonparametric method, say smoothing spline; and for sparsity in sparse linear regression.
- (2) An alternative approach is to extend the basis approach in Problem 2b. Specifically, you can assume that for the unknown function f_j , a set of basis functions $\{N_{j,l}(\cdot)\}_{1 \leq l \leq M_j}$ can be constructed to approximate $f_j(\cdot)$ accurately enough.